

Assignment 1

16.4

Example 3: Verify both forms of Green's theorem for the vector field

$$F(x, y) = (x - y)i + xj$$

and the region R bounded by the unit circle

$$C: r(t) = (\cos t)i + (\sin t)j, \quad 0 \leq t \leq 2\pi$$

$$\square F = Mi + Nj \text{ around } C$$

$$x = \cos t \text{ and } y = \sin t \text{ On the curve } C$$

$$F(r(t)) = ?$$

$$M = x - y = \cos t - \sin t$$

$$N = x = \cos t$$

$$dx = d(\cos t) = -\sin t \, dt$$

$$dy = d(\sin t) = \cos t \, dt$$

Therefore,

$$\oint_C F \cdot T \, ds = \oint_C M \, dx + N \, dy$$

$$= \int_{t=0}^{t=2\pi} (\cos t - \sin t)(-\sin t) \, dt + (\cos t)(\cos t) \, dt$$

$$= \int_0^{2\pi} (-\sin t \cos t + 1) \, dt = 2\pi$$

Since, $M = x - y$ and $N = x$

$$\frac{\partial M}{\partial x} = 1, \quad \frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = 1, \quad \frac{\partial N}{\partial y} = 0$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \iint_R (1 - (-1)) dx dy$$

$$= 2 \iint_R dx dy = 2 (\text{area inside the unit circle}) = 2\pi$$

Thus, the right and left sides of Equation both equal 2π , as asserted by the circulation flux version of Green's Theorem

$$\oint_C M dy - N dx = \int_{t=0}^{t=2\pi} (\cos t - \sin t)(\cos t dt) - (\cos t)(-\sin t dt)$$

$$= \int_0^{2\pi} \cos^2 t dt = \pi$$

$$\iint_R \left(\left(\frac{\partial M}{\partial x} \right) + \frac{\partial N}{\partial y} \right) dx dy = \iint_R (1 + 0) dx dy = \iint_R dx dy = \pi$$

Hence, the right and left sides of Equation both equal π , as asserted by the flux divergence version of Green's theorem.

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Example 3)

where C

$$0 \leq t \leq$$

$$x = \cos$$

$$\int_C -y dx$$

16.2

Example 3) Evaluate the line integral $\int_C -y dx + z dy + 2x dz$,
 where C is the helix $r(t) = (\cos t)i + (\sin t)j + tk$,
 $0 \leq t \leq 2\pi$

$$x = \cos t, \quad y = \sin t, \quad z = t, \quad \begin{aligned} dz &= dt \\ dx &= -\sin t dt \\ dy &= \cos t dt \end{aligned}$$

$$\begin{aligned} \int_C -y dx + z dy + 2x dz &= \int_0^{2\pi} [(-\sin t)(-\sin t) + t \cos t + 2 \cos t] dt \\ &= \int_0^{2\pi} [2 \cos t + t \cos t + \sin^2 t] dt \\ &= \left[2 \sin t + (t \sin t + \cos t) + \left(\frac{t}{2} - \frac{\sin 2t}{4} \right) \right]_0^{2\pi} \\ &= [0 + (0+1) + (\pi-0)] - [0 + (0+1) + (0-0)] \\ &= \pi \end{aligned}$$

Example 4) Find the work done by the Force field

$F = (y - x^2)\mathbf{i} + (z - y^2)\mathbf{j} + (x - z^2)\mathbf{k}$ in moving an object along the curve $r(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$, $0 \leq t \leq 1$, from $(0,0,0)$ to $(1,1,1)$

$$F = (y - x^2)\mathbf{i} + (z - y^2)\mathbf{j} + (x - z^2)\mathbf{k} \\ = (t^2 - t^2)\mathbf{i} + (t^3 - t^4)\mathbf{j} + (t - t^6)\mathbf{k}$$

$$\frac{dr}{dt} = \frac{d}{dt}(t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}) = \mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}$$

$$F \cdot \frac{dr}{dt} = [(t^3 - t^4)\mathbf{j} + (t - t^6)\mathbf{k}] \cdot (\mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}) \\ = t^3 - t^4(2t) + (t - t^6)(3t^2) = 2t^4 - 2t^5 + 3t^3 - 3t^8$$

So,

$$\text{Work} = \int_a^b F \cdot \frac{dr}{dt} dt = \int_0^1 (2t^4 - 2t^5 + 3t^3 - 3t^8) dt$$

$$= \left[\frac{2}{5} t^5 - \frac{2}{6} t^6 + \frac{3}{4} t^4 - \frac{3}{9} t^9 \right]_0^1 = \frac{29}{60}$$

Example 5) Find the workdone by the force field $F = xi + yj + zk$ in moving an object along the curve C parametrized by $r(t) = \cos(\pi t)i + t^2j + \sin(\pi t)k$, $0 \leq t \leq 1$

$$F(r(t)) = \cos(\pi t)i + t^2j + \sin(\pi t)k$$

$$\frac{dr}{dt} = -\pi \sin(\pi t)i + 2tj + \pi \cos(\pi t)k$$

$$\cos(\pi t) = 2t^3$$

$$F(r(t)) \cdot \frac{dr}{dt} = -\pi \sin(\pi t) \cos(\pi t) + 2t^3 + \pi \sin(\pi t)$$

The workdone is the line integral

$$\int_a^b F(r(t)) \cdot \frac{dr}{dt} dt = \int_0^1 2t^3 dt = \left[\frac{t^4}{2} \right]_0^1 = \frac{1}{2}$$

Example 6) A fluid's velocity field is $F = xi + zj + yk$. Find the flow along the helix $r(t) = (\cos t)i + (\sin t)j + tk$, $0 \leq t \leq \pi/2$

$$F = xi + zj + yk = (\cos t)i + tj + (\sin t)k$$

$$\frac{dr}{dt} = (-\sin t)i + (\cos t)j + k$$

$$F \cdot \frac{dr}{dt} = (\cos t)(-\sin t) + (t)(\cos t) + (\sin t)(1) = -\sin t \cos t + t \cos t + \sin t$$

Finally, we integrate $F \cdot (dr/dt)$ from $t=0$ to $t=\frac{\pi}{2}$

$$\text{Flow} = \int_{t=a}^{t=b} F \cdot \frac{dr}{dt} dt = \int_0^{\pi/2} (-\sin t \cos t + t \cos t + \sin t) dt$$

$$= \left[\frac{\cos^2 t}{2} + t \sin t \right]_0^{\pi/2} = \left(0 + \frac{\pi}{2} \right) - \left(\frac{1}{2} + 0 \right)$$

$$= \frac{\pi}{2} - \frac{1}{2}$$

Q.7) $F = 3yi + 2xj + 4zk$

a) The straight line path $C_1: r(t) = ti + tj + tk$
 $0 \leq t \leq 1$

$$f = 3ti + 2tj + 4tk$$

$$\frac{dr}{dt} = i + j + k$$

$$f \cdot \frac{dr}{dt} = 3t + 2t + 4t$$

$$f \cdot \frac{dr}{dt} = 9t$$

$$\int_0^1 f \cdot \frac{dr}{dt} dt = \int_0^1 9t dt$$

$$= \left| \frac{9}{2} t^2 \right|_0^1$$

$$= \frac{9}{2} - 0$$

$$\int_0^1 f \cdot \frac{dr}{dt} = \frac{9}{2}$$

Exercise 14.7)

① $x^2 + xy + y^2 + 3x - 3y + 4 = f(x, y)$

$$f_x = 2x + y + 3$$

$$f_y = 2y + x - 3$$

$$f_{xx} = 2$$

$$f_{yy} = 2$$

$$f_{xy} = 1$$

$$f_{xx} f_{yy} - (f_{xy})^2$$

$$(2)(2) - (1)^2$$

$$3 > 0$$

Since, $f_{xx} > 0$ and $f_{xx} f_{yy} - (f_{xy})^2 > 0$ therefore Minima

② $f(x, y) = 2xy - 5x^2 - 2y^2 + 4x + 4y - 4$

$$f_x = 2y + 4 - 10x$$

$$f_y = 2x + 4 - 4y$$

$$f_{xx} = -10$$

$$f_{yy} = -4$$

$$f_{xy} = 2$$

$$f_{xx} f_{yy} - (f_{xy})^2$$

$$(-10)(-4) - (2)^2$$

$$36 > 0$$

Since, $f_{xx} < 0$ and $f_{xx} f_{yy} - (f_{xy})^2 > 0$ therefore Maxima

$$\textcircled{3} f(x, y) = x^2 + xy + 3x + 2y + 5$$

$$f_x = 2x + y + 3$$

$$f_{xx} = 2$$

$$f_y = x + 2$$

$$f_{yy} = 0$$

$$f_{xy} = 1$$

$$f_{yy} f_{xx} - (f_{xy})^2$$

$$(2)(0) - 1 < 0$$

saddle point

Exercise 15.5 (Triple Integral)

$$\textcircled{1} \int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$$

$$\int_0^1 \int_0^1 \left[x^2 z + y^2 z + \frac{z^3}{3} \right]_0^1 dy dx$$

$$\int_0^1 \int_0^1 \left(x^2 + y^2 + \frac{1}{3} \right) dy dx$$

$$\int_0^1 \left[x^2 y + \frac{y^3}{3} + \frac{1}{3} y \right]_0^1 dx$$

$$z = \int_0^1 \frac{2}{3} + x^2 \, dx$$

$$= \left[\frac{2x}{3} + \frac{x^3}{3} \right]_0^1$$

$$= \left[\frac{2}{3} + \frac{1}{3} \right]$$

$$= \frac{3}{3} = 1$$

$$(2) \int_0^{\sqrt{2}} \int_0^{3y} \int_{x^2+3y^2}^{8-x^2-y^2} dz \, dx \, dy$$

$$\int_0^{\sqrt{2}} \int_0^{3y} \left[z \right]_{x^2+3y^2}^{8-x^2-y^2} dx \, dy$$

$$\int_0^{\sqrt{2}} \int_0^{3y} \left[(8-x^2-y^2) - (x^2+3y^2) \right] dx \, dy$$

$$\int_0^{\sqrt{2}} \int_0^{3y} 8 - 2x^2 - 4y^2 \, dx \, dy$$

$$= \int_0^{\sqrt{2}} \int_0^{3y} (-2x^2 - 4y^2 + 8) \, dx \, dy$$

$$= \int_0^{\sqrt{2}} \left[\frac{-2x^3}{3} - 4xy^2 + 8x \right]_0^{3y} dy$$

$$= \int_0^{\sqrt{2}} \left[\left(\frac{-2(3y)^3}{3} - 4(3y)y^2 + 8(3y) \right) - 0 \right] dy$$

$$= \int_0^{\sqrt{2}} (24y - 30y^3) \, dy$$

$$= \int_0^{\sqrt{2}} \left[\frac{24y^2}{2} - \frac{30y^4}{4} \right]_0^{\sqrt{2}}$$

$$= \frac{12(\sqrt{2})^2 - 15(\sqrt{2})^4}{2}$$

$$= 24 - 30$$

$$= -6$$

$$\textcircled{3} \int_{-1}^1 \int_0^1 \int_0^2 (x+y+z) dy dx dz$$

$$\int_{-1}^1 \int_0^1 \int_0^2 x+y+z dy dx dz$$

$$\int_{-1}^1 \int_0^1 \left[xy + \frac{y^2}{2} + zy \right]_0^2 dx dz$$

$$\int_{-1}^1 \int_0^1 \left[\left(2x + \frac{2^2}{2} + 2z \right) - (0) \right] dx dz$$

$$\int_{-1}^1 \left[\frac{2x^2}{2} + 2x + 2zx \right]_0^1 dz$$

$$\int_{-1}^1 \left[\left(\frac{2(1)^2}{2} + 2(1) + 2z \right) - 0 \right] dz$$

$$\int_{-1}^1 (2z + 3) dz$$

$$= \left[\frac{2z^2}{2} + 3z \right]_{-1}^1$$

$$= \left[\frac{2(1)^2}{2} + 3(1) \right] - \left[\frac{2(-1)^2}{2} + 3(-1) \right]$$

$$= 6$$

14-8

① Extrema on an ellipse Find the points on the ellipse $x^2 + 2y^2 = 1$ where $f(x, y) = xy$ has its extreme values

$$f(x, y) = xy$$

$$g(x, y) = x^2 + 2y^2 - 1 = 0$$

$$\nabla f = \lambda \nabla g$$

Gradient Equation:

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$$

$$= y \hat{i} + x \hat{j}$$

$$\nabla g = \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j}$$

$$= 2x \hat{i} + 4y \hat{j}$$

$$\lambda (\nabla g) = 2\lambda x \hat{i} + 4\lambda y \hat{j}$$

$$\therefore \Delta f = \lambda \nabla g$$

$$y \hat{i} + x \hat{j} = (2\lambda x) \hat{i} + (4\lambda y) \hat{j}$$

comparing $\hat{i}$ and $\hat{j}$

$$y = 2\lambda x$$

$$x = 4\lambda y$$

put x value in y :-

$$y = 2\lambda(4xy)$$

$$y = 8\lambda^2 y$$

$$\lambda^2 = \frac{y}{8y}$$

$$\lambda^2 = \frac{1}{8}$$

$$\lambda = \pm \frac{1}{2\sqrt{2}}$$

put λ value in x :-

$$x = 4\left(\pm \frac{1}{2\sqrt{2}}\right)y$$

$$x = \pm \frac{2}{\sqrt{2}}y$$

$$g(x, y) = x^2 + 2y^2 - 1 = 0$$

$$z\left(\pm \frac{2y}{\sqrt{2}}\right)^2 + 2y^2 - 1 = 0$$

$$z\left(\pm \frac{\sqrt{2}\sqrt{2}}{\sqrt{2}}\right)^2 y^2 + 2y^2 - 1 = 0$$

$$z \pm 2y^2 + 2y^2 - 1 = 0$$

$$z \pm 4y^2 - 1 = 0$$

$$z = y^2 \pm \frac{1}{4}$$

$$z = y^2 \pm \frac{1}{2}$$

put y value in x :-

$$x = \pm \frac{2}{\sqrt{2}}\left(\pm \frac{1}{2}\right)$$

$$x = \pm \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$x = \pm \frac{\sqrt{2}}{2}$$

Therefore f takes on its extreme values at $\left(\pm \frac{\sqrt{2}}{2}, \frac{1}{2}\right)$

and $\left(\pm \frac{\sqrt{2}}{2}, -\frac{1}{2}\right)$. The extreme values of f on the ellipse are $\pm \frac{\sqrt{2}}{2}$

(3) Maximum on a line Find the maximum value of $f(x,y) = 49 - x^2 - y^2$ on the line $x + 3y = 10$

$$f(x,y) = 49 - x^2 - y^2$$

$$g(x,y) = x + 3y - 10 = 0$$

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$$

$$\nabla f = -2x \hat{i} - 2y \hat{j}$$

$$\nabla g = \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j}$$

$$\nabla g = \hat{i} + 3\hat{j}$$

$$\nabla f = \lambda (\nabla g)$$

$$(-2x)\hat{i} + (-2y)\hat{j} = \lambda (\hat{i} + 3\hat{j})$$

$$(-2x)\hat{i} + (-2y)\hat{j} = \lambda (\hat{i} + 3\hat{j})$$

Comparing

$$\lambda = -2x$$

$$\lambda = \frac{-2x}{2}$$

$$3\lambda = -2y$$

$$y = \frac{-3\lambda}{2}$$

of $f(x, y) = 49 -$

$$g(x, y) = x + 3y - 10 = 0$$

$$z \frac{-\lambda}{2} + 3 \left(\frac{-3}{2} \right) \lambda - 10 = 0$$

$$z \frac{-\lambda}{2} - \frac{9\lambda}{2} - 10 = 0$$

$$\frac{z(-\lambda - 9\lambda - 20)}{2} = 0$$

$$-10\lambda - 20 = 0$$

$$z - 10(\lambda + 2) = 0$$

$$z \lambda = -2$$

put λ value in x :

$$x = \frac{-\lambda}{2} = \frac{-(-2)}{2} = 1$$

put λ value in y :

$$y = \frac{-3}{2} \lambda = \frac{-3}{2} (-2) = 3$$

values of x, y and λ

$$f(x, y) = 49 - x^2 - y^2$$

$$= 49 - 1^2 - 3^2$$

$$= 39$$

⑤ Constrained minimum. Find the points on the curve $xy^2 = 54$ nearest the origin.

$$f(x, y) = x^2 + y^2$$

$$g(x, y) = xy^2 = 54$$

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \\ &= 2x \hat{i} + 2y \hat{j}\end{aligned}$$

$$\nabla g = y^2 \hat{i} + 2xy \hat{j}$$

$$\nabla f = \lambda \nabla g$$

$$2x \hat{i} + 2y \hat{j} = \lambda (y^2 \hat{i} + 2xy \hat{j})$$

comparing

$$\begin{aligned}2x &= \lambda y^2 \\ y^2 &= \frac{2x}{\lambda}\end{aligned}$$

$$\begin{aligned}2y &= 2\lambda xy \\ x &= \frac{1}{\lambda}\end{aligned}$$

put x value in y

$$y^2 = \frac{2}{\lambda} \left(-\frac{1}{\lambda}\right)$$

$$y^2 = \frac{2}{\lambda^2}$$

put in question

$$xy^2 = 54$$

$$\left(\frac{1}{\lambda}\right) \left(\frac{2}{\lambda^2}\right) = 54$$

$$\frac{2}{\lambda^3} = 54$$

$$\lambda^3 = \frac{1}{27} = \lambda^3 = \frac{1}{3^3} \quad \lambda = \frac{1}{3}$$

put λ in x and y

$$x = \frac{1}{\lambda} = \frac{1}{1/3} = 3$$

$$y^2 = \frac{2}{\lambda^2} = \frac{2}{(1/3)^2} = 18$$

$$y = \pm 3\sqrt{2}$$

$$(3, 3\sqrt{2}), (3, -3\sqrt{2})$$

Assignment 2

Stokes Theorem

Solved Example ppt

Use Stokes's theorem to calculate the circulation of the field

$F = x^2\mathbf{i} + 2x\mathbf{j} + z^2\mathbf{k}$ around the curve C = The ellipse

$4x^2 + y^2 = 4$ in the xy plane counter when viewed from above.

The surface S is $4x^2 + y^2 = 4$ i.e. $F = 4x^2 + y^2 = 4$

$$\nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 2x & z^2 \end{vmatrix}$$

$$(\nabla \times F) \cdot \mathbf{k} = 2$$

$$\text{Circulation } \oint_C F \cdot d\mathbf{r} = \int_0^{2\pi} \int_{\frac{-\sqrt{4-y^2}}{2}}^{\frac{\sqrt{4-y^2}}{2}} 2 \, dx \, dy$$

$$= 2 \int_0^2 \sqrt{4-y^2} \, dy$$

$$= 2 \left(\frac{y}{2} \sqrt{4-y^2} + \frac{4}{2} \sin^{-1} \frac{y}{2} \right) \Big|_0^2$$

$$= 4 \left(\frac{\pi}{2} - 0 \right) = 2\pi$$

Solved Example ppt

The Divergence Theorem

The flux of a vector field $F = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ across a closed oriented surface S in the direction of the surface's outward unit normal field $\mathbf{n}$ equals the integral of $\nabla \cdot F$ (divergence of F) over the region D enclosed by the surface:

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_D \nabla \cdot \mathbf{F} \, dV$$

$$\mathbf{F} = y\mathbf{i} + xy\mathbf{j} - z\mathbf{k}$$

D : The region inside the solid cylinder $x^2 + y^2 \leq 4$ between the plane $z = 0$ and paraboloid $z = x^2 + y^2$

$$\nabla \cdot \mathbf{F} = 0 + x - 1 = x - 1$$

$$\iiint_D \nabla \cdot \mathbf{F} \, dV = \int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{x^2+y^2} (x-1) \, dz \, dy \, dx$$

$$= \int_0^2 (x-1) \left[x^2 y + \frac{y^3}{3} \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx$$

$$= \int_0^2 (x-1) \left(2x^2 \sqrt{4-x^2} + \frac{2}{3} (4-x^2)^2 \sqrt{4-x^2} \right) dx$$

$$= \frac{1}{3} \int_0^2 (x-1) \sqrt{4-x^2} [6x^2 + 2(16-8x+8x^2)] dx$$

$$= \frac{1}{3} \int_0^2 (x-1) \sqrt{4-x^2} [8x^2 - 8x + 16] dx$$

$$= -16\pi$$

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Example 3) Find the flux of $F = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$ outward through the surface of the cube cut from the first octant by the planes $x=1, y=1$ and $z=1$

$$\nabla \cdot F = \frac{\partial}{\partial x} (xy) + \frac{\partial}{\partial y} (yz) + \frac{\partial}{\partial z} (xz)$$

$$= y + z + x$$

Over the cube's interior:

$$\text{Flux} = \iint_{\text{cube surface}} F \cdot \mathbf{n} \, d\sigma = \iiint_{\text{cube interior}} \nabla \cdot F \, dv$$

$$= \int_0^1 \int_0^1 \int_0^1 (x+y+z) \, dx \, dy \, dz$$

$$= \frac{3}{2}$$

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Example 1) Find the parametrization of the cone

$$z = \sqrt{x^2 + y^2}, \quad 0 \leq z \leq 1$$

In cylindrical coordinates:

- $x = r \cos \theta$
- $y = r \sin \theta$
- $z = r$

Thus, the cone's equation becomes $z = \sqrt{x^2 + y^2}$, which simplifies to $z = r$

Hence, the parametrization is:

$$\mathbf{r}(r, \theta) = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j} + r \mathbf{k} \quad \text{with } 0 \leq r \leq 1 \text{ and } 0 \leq \theta \leq 2\pi$$

Example 2) Find the parametrization of the sphere

$$x^2 + y^2 + z^2 = a^2$$

In spherical coordinates:

- $x = a \sin \phi \cos \theta$
- $y = a \sin \phi \sin \theta$
- $z = a \cos \phi$

Thus,

Thus, the parametrization is

$$r(\phi, \theta) = a \sin \phi \cos \theta \mathbf{i} + a \sin \phi \sin \theta \mathbf{j} + a \cos \phi \mathbf{k}$$

with $0 \leq \phi \leq \pi$ and $0 \leq \theta \leq 2\pi$

Example 3) Find a parametrization of the cylinder

$$x^2 + (y-3)^2 = 9 \quad 0 \leq z \leq 5$$

In the cylindrical coordinates:

- $x = r \cos \theta$

- $y = r \sin \theta + 3$

- $z = z$

For the cylinder:

$$x^2 + (y-3)^2 = 9$$

This is a cylinder of radius 3 centred at $y=3$

Thus, $r=3$ and the parametrization is:

$$r(\theta, z) = 3 \cos \theta \mathbf{i} + (3 \sin \theta + 3) \mathbf{j} + z \mathbf{k}$$

with $0 \leq \theta \leq 2\pi$ and $0 \leq z \leq 5$

EX 14.6

$$Q.1) x^2 + y^2 + z^2 = 3 \quad \begin{matrix} x_0 & y_0 & z_0 \\ P_0(1, 1, 1) \end{matrix}$$

$$f_x = 2x \quad \text{at } (1, 1, 1) = 2$$

$$f_y = 2y \quad \text{at } (1, 1, 1) = 2$$

$$f_z = 2z \quad \text{at } (1, 1, 1) = 2$$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$2(x-1) + 2(y-1) + 2(z-1) = 0$$

$$2x + 2y + 2z - 6 = 0$$

Normal line

$$x = 1 + 2t, \quad y = 1 + 2t, \quad z = 1 + 2t$$

$$Q.2) x^2 + y^2 - z^2 = 18 \quad P_0(3, 5, -4)$$

$$f_x = 2x \quad \text{at } (3, 5, -4) = 6$$

$$f_y = 2y \quad \text{at } (3, 5, -4) = 10$$

$$f_z = -2z \quad \text{at } (3, 5, -4) = 8$$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$6(x-3) + 10(y-5) + 8(z-(-4)) = 0$$

$$6x - 18 + 10y - 50 + 8z + 32 = 0$$

$$6x + 10y + 8z - 36 = 0$$

$$x = 3 + 6t$$

$$y = 5 + 10t$$

$$z = -4 + 8t$$

$$(3) \quad 2z - x^2 = 0 \quad P_0(2, 0, 2)$$

$$f_x = -2x \quad \text{at } (2, 0, 2) = -4$$

$$f_y = 0 \quad \text{at } (2, 0, 2) = 0$$

$$f_z = 2 \quad \text{at}$$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$-2x(x-2) + 0(y-0) + 2(z-2) = 0$$

$$-2x^2 + 4x + 2z - 4 = 0$$

$$-4(x-2) + 0(y-0) + 2(z-2) = 0$$

$$-4x + 8 + 2z - 4 = 0$$

$$-4x + 4 + 2z = 0$$

$$x = 2 - 4t$$

$$y = 0$$

$$z = 2 + 2t$$

$$(4) \quad x^2 + 2xy - y^2 + z^2 = 7 \quad P_0(1, -1, 3)$$

$$f_x = 2x + 2y \quad \text{at } (1, -1, 3) = 0$$

$$f_y = -2y + 2x \quad \text{at } (1, -1, 3) = 4$$

$$f_z = 2z \quad \text{at } (1, -1, 3) = 6$$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$0(x-1) + 4(y-(-1)) + 6(z-3) = 0$$

$$4y + 4 + 6z - 18 = 0$$

$$4y + 6z - 14 = 0$$

$$x = 1$$

$$y = -1 + 4t$$

$$z = 3 + 6t$$

$$(5) \cos \pi x - x^2 y + e^{xz} + yz = 4 \quad P_0(0, 1, 2)$$

$$f_x = -\sin \pi x (\pi) - 2xy + e^{xz} \times z$$

$$f_x = -\pi \sin \pi x - 2xy + e^{xz} z = 2$$

$$f_y = -x^2 + z \quad \text{at } (0, 1, 2) = 2$$

$$f_z = y + e^{xz} x \quad \text{at } (0, 1, 2) = 1$$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$2(x-0) + 2(y-1) + 1(z-2) = 0$$

$$2x + 2y - 2 + z - 2 = 0$$

$$2x + 2y + z - 4 = 0$$

$$x = 2t$$

$$y = 1 + 2t$$

$$z = 2 + t$$

$$\textcircled{6} \quad x^2 + 2xy - y^2 + z$$

$$x^2 - xy - y^2 - z = 0 \quad P_0(1, 1, -1)$$

$$f_x = 2x - y \quad \text{at } (1, 1, -1) = 1$$

$$f_y = -2y - x \quad \text{at } (1, 1, -1) = -3$$

$$f_z = -1 \quad \text{at } (1, 1, -1) = -1$$

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0$$

$$1(x-1) - 3(y-1) - 1(z-(-1)) = 0$$

$$x - 1 - 3y + 3 - z - 1 = 0$$

$$x - 3y + 1 - z = 0$$

$$x = 1 + 1t$$

$$y = 1 - 3t$$

$$z = -1 - 1t$$

Multi variable Partial Derivative

$$\text{Q.1) } f(x, y) = 2x^2 - 3y - 4$$

$$\frac{\partial f}{\partial x} = 4x$$

$$\frac{\partial f}{\partial y} = -3$$

$$\textcircled{2} \quad f(x, y) = x^2 - xy + y^2$$

$$\frac{\partial f}{\partial x} = 2x - y$$

$$\frac{\partial f}{\partial y} = 2y - x$$

$$\textcircled{3} \quad f(x, y) = e^{(x+y+1)}$$

$$\frac{\partial f}{\partial x} = e^{(x+y+1)} \cdot (1)$$

$$= e^{(x+y+1)}$$

$$\frac{\partial f}{\partial y} = e^{(x+y+1)} \cdot (1)$$

$$= e^{(x+y+1)}$$